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Nonbank Fragility in Credit Markets: Evidence from a Two-Layer Asset Demand System

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1 Big picture

- **Question.** How does nonbank fragility arise in corporate bond markets, and how much can household flows into and out of bond funds amplify shocks to asset prices?
- **Setting.** The paper studies the U.S. corporate bond market, with special attention to the March 2020 COVID turmoil and, as a second application, the 2015–2016 energy episode.
- **Core challenge.** Asset prices and fund flows are jointly determined in equilibrium. When prices fall, mutual funds lose value and face redemptions; those redemptions force sales that push prices down further. Measuring this amplification requires a framework that combines fund flows, institutional asset demand, and overlapping holdings across many assets and investor sectors.
- **Main conceptual contribution.** The paper develops a *two-layer asset demand system.* In the first layer, households allocate wealth across institutions. In the second layer, institutions allocate wealth across specific assets. This delivers a tractable model with:
  - price–flow feedback loops,
  - contagion across assets,
  - contagion across institutions,
  - and policy counterfactuals that can be tied directly to estimated sufficient statistics.
- **Main sufficient statistics.** Everything is organized around a few estimated objects:
  - flow-to-performance sensitivities in the mutual fund sector,
  - demand elasticities of different investor groups,
  - and the distribution of asset holdings across those groups.
- **Data and estimation.** The flow layer is estimated from CRSP and Morningstar mutual fund data. The asset-demand layer is estimated from bond prices, bond characteristics, and holdings data from WRDS Bond Returns, FISD, eMAXX, and CRSP.
- **Main baseline result.** The model implies sizable amplification in the March 2020 crisis. Negative shocks to bond values trigger outflows from mutual funds, and these outflows create additional price pressure on both the directly affected assets and on other assets that funds sell to meet redemptions.
- **Main cross-asset result.** All four broad bond classes in the paper's calibration are systemic, but long high-yield bonds are the most systemic. An especially striking finding is that short investment-grade bonds are *more* systemic than long investment-grade bonds, because they are disproportionately held by flow-sensitive mutual funds.
- **Main institutional result.** Insurers are more price-elastic and do not face mutual-fund-style demandable liabilities, so they act as a stabilizing force relative to mutual funds. By contrast, active mutual funds are inelastic and flow-sensitive. The residual sector is also far from infinitely elastic, which means slow-moving arbitrage capital does not quickly eliminate dislocations.
- **Main policy result.** Conventional rate cuts, corporate bond purchases, direct financing of mutual funds, redemption restrictions, and swing pricing all help, but their effectiveness differs sharply. Direct financing of mutual funds and broad asset purchases have the highest "bang for the buck."

- **Targeting result.** A policy does not need to focus on the assets that fall the most or on the institutions with the biggest outflows. The best-targeted intervention is the one that attacks the nodes in the network that generate the most amplification.
- **Economic takeaway.** Nonbank fragility in credit markets is not just about price illiquidity in a single asset. It is about a whole system in which household reallocations, institutional mandates, and inelastic asset demand interact. The paper turns that idea into an empirically estimable framework.

2 Data and measurement (key objects)

2.1 Bond universe and institutional coverage

- The bond sample includes U.S. dollar corporate bonds issued by nonfinancial, nonutility, nonsovereign firms with issuance size above $100 million.
- The paper excludes convertibles, PIK securities, capital impact bonds, community investment bonds, and exchange issues.
- Holdings data span 2007–2021 and are merged from Thomson Reuters eMAXX and CRSP mutual fund holdings.
- Pricing data come from WRDS Bond Returns; bond characteristics come from Mergent FISD.
- The estimation sample contains 3,602 insurers, 2,967 funds, and 17,322 unique corporate bonds.

Interpretation. The paper uses broad coverage of both asset holdings and asset prices so it can estimate not just price effects, but who holds what, who is flow-sensitive, and how those holdings map into systemic fragility.

2.2 Flow data and institution groups

- Flow-to-performance estimates are built from a monthly panel of fixed-income mutual funds from January 1992 to December 2021.
- The paper splits investors into six sectors for the baseline crisis simulations:
  - life insurers,
  - property & casualty insurers,
  - active mutual funds,
  - index mutual funds,
  - ETFs,
  - and a residual sector.
- The residual sector combines observed non-mutual-fund and non-insurer holders plus a final residual holder that absorbs the remaining outstanding amount of each bond.

Interpretation. This grouping is central: the paper does not treat “the market” as a representative investor. Fragility depends on which investors hold each asset and how elastic or flow-sensitive those investors are.

2.3 Asset classes versus bond types

The paper uses two levels of aggregation.

- **Bond types** are used for demand estimation. There are 72 types based on:
  - credit rating: A, BBB, or high-yield;
  - remaining maturity: up to 3 years, 3–10 years, 10+ years;
  - issuance size: below or above $500 million;
  - covenant-lite status;
  - redemption option.
- **Asset classes** are used for the crisis simulations:
  - investment-grade long,
  - investment-grade short,
  - high-yield long,
  - high-yield short.

Interpretation. The bond-type aggregation is an econometric device that makes the IV demand estimation feasible in the corporate bond market, where there are far too many individual securities with sparse holdings. The four asset classes are a higher-level summary used for transparency in the crisis and policy exercises.

2.4 Fund flow measurement

Net flow is defined as asset growth adjusted for fund returns:

$$Flow_{i,t} = \frac{TNA_{i,t} - TNA_{i,t-1}(1 + v_{i,t})}{TNA_{i,t-1}}, \tag{1}$$

where $TNA_{i,t}$ is total net assets and $v_{i,t}$ is the fund return over the prior month.

Interpretation. This is standard flow measurement. It isolates true investor inflows and outflows from the mechanical change in fund size caused by asset returns.

2.5 Representative-household simplification

In the first layer of the model, the general household-by-institution flow matrix is collapsed by assuming a representative household in the empirical implementation. That means the object to estimate is institution-level flow sensitivity rather than household-specific sensitivity.

Interpretation. This simplification keeps the flow layer tractable and lets the paper map mutual fund performance into mutual fund flows directly.

2.6 Demand-estimation outcome

For institution i, bond type n, and period t, the paper estimates demand using the change in holdings *relative* to a benchmark index bond fund:

$$\Delta\tilde{q}_{it}(n) = \Delta q_{it}(n) - \Delta q_{It}(n), \tag{2}$$

where $\Delta q_{it}(n)$ is the change in holdings of bond type n normalized by institution i's assets under management, and I denotes the largest index bond fund used as the benchmark.

Interpretation. This is very important. It nets out the mechanical trading created by index rebalancing, so the estimated elasticity captures active price-sensitive trading rather than passive benchmark maintenance.

2.7 Instrument for demand estimation

The key IV is built from the idiosyncratic component of fund flows of *other* institutions that held a given bond type in the prior period:

$$z_{it}^{flow}(k) = \frac{\sum_{j\neq i} \hat{f}_{j,t}^{\perp}\, AUM_{j,t-1}\, w_{j,t-1}(k)}{mc_{k,t-1}}, \tag{3}$$

where $\hat{f}_{j,t}^{\perp}$ is the orthogonalized flow of institution j, $w_{j,t-1}(k)$ is its lagged portfolio weight in bond type k, and $mc_{k,t-1}$ is the lagged market capitalization of that bond type.

Interpretation. The idea is to use exogenous shifts in the demand coming from other holders of the same bond type to instrument for price changes.

2.8 Controls in the demand estimation

The demand estimation controls for:

- duration-matched Treasury yields,
- issuer credit rating,
- years remaining to maturity,
- amount issued,
- and bid–ask spreads.

It also includes bond-type fixed effects and fund-by-year fixed effects.

Interpretation. These controls help isolate price-sensitive portfolio rebalancing from simple changes in observable bond characteristics or time-varying fund mandates.

2.9 Descriptive facts emphasized by the paper

- Across mutual funds, flows are highly sensitive to returns, especially bad returns.
- At the end of 2019, mutual funds and ETFs together held roughly 20% of the corporate bond market.

- Hedge funds held only a very small share of corporate bonds, and their holdings actually fell in early 2020.

Interpretation. This is why nonbank fragility is quantitatively important: a meaningful share of the market is held by investors with demandable liabilities, while the "arbitrage" sector is too small and too slow to offset them.

3 Models and empirical specifications (all equations + interpretation)

3.1 Layer 1: household demand for institutions

Households choose among institutions to maximize indirect utility:

$$\max_{i \in I} u_{h,i,t} = \beta_{h,i} X_{i,t} + \epsilon_{h,i,t}. \tag{4}$$

This yields logit portfolio weights:

$$\theta_{h,i,t} = \frac{\exp(\beta_{h,i} X_{i,t})}{\sum_{i=0}^{I} \exp(\beta_{h,i} X_{i,t})}. \tag{5}$$

Demand for institution liabilities is:

$$Q^D_{h,i,t} = \frac{\theta_{h,i,t} W^*_{h,t}}{V^*_{i,t}}. \tag{6}$$

Interpretation. The first layer determines institution size. High fund returns raise fund attractiveness, which raises household demand for that institution and therefore its AUM.

3.2 Layer 2: institution demand for assets

Institution i chooses among assets to maximize:

$$\max_{n \in N} u_{i,t}(n) = \kappa_{i,n} X_{n,t} + \epsilon_{i,n,t}. \tag{7}$$

This implies logit portfolio weights:

$$\theta_{i,n,t} = \frac{\exp(\kappa_{i,n} X_{n,t})}{\sum_{n=0}^{N} \exp(\kappa_{i,n} X_{n,t})}. \tag{8}$$

Institution NAV is:

$$V_{i,t} = \sum_{n=0}^{N} \theta_{i,n,t} P_{n,t}. \tag{9}$$

Demand for assets is:

$$Q^D_{i,n,t} = \frac{\theta_{i,n,t} W^*_{i,t}}{P^*_{n,t}}. \tag{10}$$

Interpretation. The second layer determines how institution-level AUM is mapped into asset-level demand. Once flows change institution size, asset demand changes mechanically through mandates and portfolio weights.

3.3 Market clearing

Institution liabilities clear when household demand equals institution share supply:

$$\sum_{h=0}^{H} Q^D_{h,i,t} = Q^S_{i,t}. \tag{11}$$

Asset markets clear when total institutional demand equals asset supply:

$$\sum_{i=0}^{I} Q^D_{i,n,t} = Q^S_{n,t}. \tag{12}$$

Interpretation. Institution shares adjust primarily through quantities because funds can create or destroy shares. Corporate bond markets adjust primarily through prices because outstanding bond quantities are fixed in the short run.

3.4 Flow-to-performance relationship

After log-linearization, the fund-flow system is:

$$f_t = B_t(\beta)\, v_t, \tag{13}$$

where f_t is the vector of institution flows and v_t is the vector of institution returns.

The flow-to-performance matrix is:

$$B_t(\beta) = \mathrm{diag}\big(1^\top (S_{H,t} \odot \beta)\big) - S^\top_{H,t}(\theta_{H,t} \odot \beta). \tag{14}$$

Interpretation. B summarizes how returns map into flows. When B is large, investors are flightier and the system is more fragile.

3.5 Institution demand for assets after log-linearization

The paper's log-linear asset-demand equation is:

$$q_t = K_t(\kappa)\, r^e_t + S^\top_t\, f_t, \tag{15}$$

where q_t is aggregate asset demand, r^e_t is expected return, and S_t maps institutions into the asset classes they hold.

The macro demand-sensitivity matrix is:

$$K_t(\kappa) = \mathrm{diag}\big(1^\top (S_t \odot \kappa)\big) - S^\top_t(\theta_t \odot \kappa). \tag{16}$$

Interpretation. K summarizes how price changes translate into demand changes. When K is small, demand is inelastic and shocks are amplified more strongly.

3.6 Pricing equation and amplification matrix

Using the expected return identity and market clearing, the pricing equation becomes:

$$p_t = \left(I + \delta A^{-1}\right)^{-1} (\delta d_t + p_{t+1}) \,, \tag{17}$$

where d_t is the cash-flow shock and A is the amplification matrix:

$$A = \left(I - (K\delta)^{-1} S^\top B\theta\right)^{-1} . \tag{18}$$

For a one-time permanent shock, the overall price effect is:

$$p_t = Ad. \tag{19}$$

Interpretation.

- If there are no flow effects ($B = 0$), there is no amplification.
- If demand is perfectly elastic ($K \to \infty$), there is no amplification.
- Amplification requires both inelastic demand and flow-sensitive investors.

3.7 Economic meaning of amplification and contagion

The model delivers three channels:

- **Amplification within an asset class:** falling prices trigger outflows, which trigger further selling of the same assets.
- **Contagion across assets:** redemptions force funds to sell assets not directly hit by the original shock.
- **Contagion across institutions:** insurers and other sectors see mark-to-market losses when mutual fund fire sales depress prices.

Interpretation. This is why the model is more than a reduced-form fire-sale regression. It explicitly traces how a shock propagates through a network of investors and overlapping portfolios.

3.8 Flow estimation

The empirical flow regression is:

$$f_{i,t} = \beta^+ v_{i,t}^+ + \beta^- v_{i,t}^- + \gamma X_{i,t} + \tau_i + \tau_t + \epsilon_{i,t}. \tag{20}$$

Interpretation. Positive and negative returns are allowed to have different effects. This matters because outflows in bad times are stronger than inflows in good times.

3.9 Demand estimation

The IV demand equation is:

$$\Delta\tilde{q}_{it}(n) = \gamma_{0i}\Delta p_t(n) + \gamma_{1i}x_t(n) + \alpha_{i,t} + \alpha_n + \nu_{it}(n). \tag{21}$$

Interpretation. The coefficient on instrumented price changes maps directly into institution-specific demand elasticity after adjusting for the portfolio weights used to translate quantity changes into price elasticities.

3.10 Cross-fund portfolio allocation and fragility

To relate portfolio choice to flow sensitivity, the paper estimates:

$$\omega_{it}(n) = \gamma_1 \hat{\beta}_i + \gamma_2 Retail_{it} + \gamma_3 \ln(TNA)_{it} + \alpha_t + \epsilon_{it}. \tag{22}$$

Interpretation. This shows whether flightier funds tilt toward more liquid assets. The answer is yes: higher-β funds hold more short-duration IG bonds and fewer long-duration bonds.

3.11 Amplification statistic used in the crisis simulations

The paper summarizes amplification during the COVID episode as:

$$Amplification = \frac{\text{Price drop with amplification} - \text{Price drop without amplification}}{\text{Price drop with amplification}}. \tag{23}$$

Interpretation. This is the share of the observed price decline attributable to the endogenous feedback loop rather than to the original fundamental shock alone.

3.12 Systemic risk across assets

The paper defines an asset systemic measure that asks: if one asset experiences a shock, how much does the aggregate bond market fall once the full amplification matrix is taken into account? In notation, the market impact of asset n is based on the nth column of A and is normalized by the market share α_n of that asset.

Interpretation. A value of 1 is the no-amplification benchmark. Values above 1 mean that a shock to that asset causes larger market-wide losses because of the network of holdings and flows.

3.13 Policy multiplier

For a government purchase vector g, the paper defines the price-impact multiplier as:

$$Policy\ multiplier(g) = \frac{\alpha^\top A(K\delta)^{-1} g}{\alpha^\top g}. \tag{24}$$

Interpretation. This is the "bang for the buck" of an intervention. It compares the aggregate price support achieved to the size of the intervention.

4 Identification and instruments (what makes the estimates credible)

4.1 Flow estimation: what identifies β

A key concern is reverse causality: a fund-specific flow shock could itself move the returns of the assets held by the fund, which would mechanically contaminate the flow-to-performance regression.

The paper's answer is that once fund fixed effects and time fixed effects are included, the identifying variation is the idiosyncratic relation between a fund's return and its own net flow within a month, after removing aggregate shocks.

Interpretation. The flow regressions are not trying to isolate a structural household demand system at the micro level. They are trying to recover the total reduced-form sensitivity of fund flows to fund performance, which is exactly the object needed for the amplification calculation.

4.2 Demand estimation: why price endogeneity is serious

Price changes and portfolio rebalancing are jointly determined in equilibrium. A naive regression of quantity changes on prices would mix:

- true downward-sloping demand,
- common shocks to expected returns,
- and mechanical trades linked to index rebalancing.

Interpretation. Without an instrument, the price coefficient is not credible.

4.3 Instrument design

The identification strategy follows the spirit of van der Beck's demand-based asset pricing approach:

- first, isolate the idiosyncratic component of flows to each institution;
- second, aggregate the leave-one-out flows of other institutions that held a bond type;
- third, use that as a source of exogenous demand pressure on the bond type.

Interpretation. The instrument shifts prices because other investors in the same bond type receive idiosyncratic flow shocks, but these flow shocks are plausibly orthogonal to the focal institution's own latent demand.

4.4 Why the benchmark-index adjustment matters

The paper subtracts changes in holdings of a benchmark index bond fund from each institution's holdings change. This is meant to strip out the very large amount of mechanical trading that comes from index maintenance and benchmark rebalancing in the bond market.

Interpretation. This is a crucial identification step. Otherwise, the paper might mistake benchmark-driven turnover for active price-sensitive demand.

4.5 Why bond-type aggregation matters

The corporate bond market has too many securities and too many zeros at the individual bond level. Aggregating into 72 bond types solves two problems at once:

- it makes the data dense enough for estimation;
- it makes the estimated elasticity relevant for substitution across economically similar securities rather than for a single individual CUSIP.

Interpretation. The estimated elasticities are therefore meaningful for crisis simulations, where the relevant margin is broad reallocations across bond categories.

4.6 Instrument strength

The first-stage coefficients are large and strongly significant across all investor groups, with coefficients on the flow instrument between roughly -1.1 and -1.4 in Table 2.

Interpretation. Exogenous demand pressure from other holders meaningfully moves spreads, so the IV is not weak.

4.7 What the paper can and cannot distinguish

- The framework can in principle study both cash-flow shocks and exogenous flow shocks.
- In the main crisis application, the paper deliberately focuses on fundamental bond-value shocks and lets mutual fund outflows emerge endogenously.
- The paper does *not* attempt to model all dimensions of policy, such as signaling effects or conditional promises by the central bank.
- The counterfactuals also hold the estimated parameters fixed over the short crisis horizon.

Interpretation. The paper is very strong on the equilibrium mechanics of flows and price pressure, but it is not a fully general model of all macro-financial policy channels.

4.8 Relation to the literature

The paper sits at the intersection of:

- asset demand system pricing,
- mutual fund fragility and runs,
- fire sales and bond market illiquidity,
- and macroprudential / unconventional policy design.

Interpretation. Its distinctive contribution is to integrate household flows and institutional asset demand in one empirically estimable system.

Table 1: Flow to performance estimates across mutual funds groups

	(1) Flow	(2) Flow	(3) Flow	(4) Flow
Positive return	0.146***	0.136***	0.028	0.218***
	[0.021]	[0.022]	[0.182]	[0.059]
Negative return	0.432***	0.442***	-0.003	0.464***
	[0.023]	[0.025]	[0.199]	[0.067]
L.Flow	0.261***	0.274***	0.224***	0.203***
	[0.003]	[0.003]	[0.019]	[0.007]
Sample	All funds	Active	Index	ETF
Fund F.E.	✓	✓	✓	✓
Time F.E.	✓	✓	✓	✓
Observations	231,113	196,026	5,969	29,105
Adj. R-squared	0.188	0.193	0.192	0.152

Note: This table shows the relationship between fund flows and returns. The sample period is from 2007 to 2021 with monthly observations. "Return" is the monthly return of the fund in percentage points. The dependent variable is the fund flow, measured by the percentage change in the asset under management from the previous month. Data source: CRSP Mutual Fund Database.

5 Tables and figures

5.1 Table 1: flow-to-performance estimates across fund groups

Read:

- For all funds, the coefficient on positive returns is 0.146 and the coefficient on negative returns is 0.432.
- For active funds, the corresponding coefficients are 0.136 and 0.442.
- For ETFs, they are 0.218 and 0.464.
- For index funds, the coefficients are small and statistically weak.

Economic message. Flows react much more strongly to bad returns than to good returns. This asymmetry is one of the main ingredients of fragility in the model.

5.2 Tables 2 and 3: IV demand estimates

Read:

- The first-stage coefficient on the flow instrument is strongly negative and highly significant for every group, ranging from about -1.12 to -1.42.
- The second-stage coefficient on term-adjusted spread changes is 1.833 for life insurers, 1.178 for P&C insurers, 0.854 for active mutual funds, and 1.031 for the residual sector.
- ETFs and index funds are close to price-inelastic in the second stage.
- After translating coefficients into elasticities, insurers are the most elastic among the main investor groups, active mutual funds have an elasticity around 0.75, and the residual sector around 1.1.

Economic message. Corporate bond demand is downward sloping and far from infinitely elastic. This matters enormously: if the residual sector or arbitrageurs were infinitely elastic, there would be much less amplification.

Table 2: IV first stage estimates

	(1) Life ins	(2) PC Ins	(3) ETF	(4) Index funds	(5) Active MFs	(6) Residual
z_it^{flow}	-1.252***	-1.149***	-1.116***	-1.215***	-1.423***	-1.376***
	(0.00990)	(0.00899)	(0.0340)	(0.0692)	(0.0115)	(0.0608)
UST × yrs	-0.0674***	-0.0588***	-0.0641***	-0.0985***	-0.0774***	-0.149***
	(0.000940)	(0.00116)	(0.00518)	(0.00759)	(0.00134)	(0.00986)
Bidask	8.528***	8.111***	13.23***	12.54***	10.81***	5.824***
	(0.0631)	(0.0630)	(0.325)	(0.675)	(0.0911)	(0.195)
Yrs remaining	0.00656***	0.00713***	0.00174***	0.00612***	0.00266***	0.0145***
	(0.000136)	(0.000241)	(0.000583)	(0.000650)	(0.000184)	(0.00115)
Amount issued (log)	-0.0000645	-0.0000781	-0.00189***	-0.00247***	-0.000156	-0.00115***
	(0.0000935)	(0.0000967)	(0.000365)	(0.000403)	(0.000116)	(0.000427)
Issuer rating	0.00124***	0.000550***	0.0000381	-0.00353***	-0.0000293	0.00123***
	(0.000115)	(0.000120)	(0.000347)	(0.000883)	(0.0000938)	(0.000425)
Constant	-0.0905***	-0.0719***	-0.0305***	-0.0337***	-0.0509***	-0.108***
	(0.00176)	(0.00197)	(0.00613)	(0.00927)	(0.00246)	(0.00845)
Bond category FE	✓	✓	✓	✓	✓	✓
Fund x Year FE	✓	✓	✓	✓	✓	✓
Observations	1474793	1272599	114047	40908	1345830	138567
R-squared	0.285	0.306	0.276	0.268	0.246	0.392

Note: This table shows the first stage estimates of the instrument on term-adjusted credit spreads within fund-bond type-quarter. The instrument is constructed from equation (25). The outcome variable in the first stage regressions is credit spread multiplied by the number of years remaining on the asset. Credit spreads are from the WRDS Bond Returns month-end transactions data, reported at the bond-quarter level. Controls include duration-matched US Treasury yield multiplied by the number of years remaining on the bond, the bid–ask spread as reported by WRDS, the number of years remaining, the initial amount issued (logged), and the issuer credit rating as reported in WRDS. The sample period is from 2007 to 2021. The first column reports the first-stage results for all life insurers, the second column reports results for property and casualty insurers, the third column reports results for ETFs, the fourth column reports results for all index funds, the fifth column reports all active mutual funds, and the last column reports results for the residual investor. Index funds include only pure index funds whose objective is to match the investment performance of a specific securities market index. Includes bond type and fund–year fixed effects. Standard errors are clustered at the fund level.

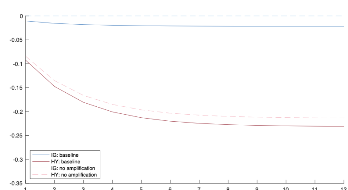
Table 3: IV second stage estimates

	(1) Life ins	(2) PC Ins	(3) ETF	(4) Index MFs	(5) Active MFs	(6) Residual
Delta CS × yrs	1.833***	1.178***	-0.267	0.636	0.854***	1.031*
	(0.228)	(0.233)	(0.324)	(0.664)	(0.0951)	(0.606)
UST × yrs	0.123***	0.0848***	0.00537	0.0132	-0.0328***	-0.0157
	(0.0193)	(0.0179)	(0.0328)	(0.0591)	(0.0107)	(0.101)
Bidask	-19.26***	-16.50***	12.97***	-14.93*	-0.873	-11.20***
	(1.811)	(1.711)	(4.829)	(8.618)	(1.125)	(3.854)
Yrs remaining	-0.00767***	-0.0131***	-0.0112***	0.00190	-0.00801***	-0.00194
	(0.00182)	(0.00244)	(0.00397)	(0.00519)	(0.000890)	(0.00887)
Amount issued (log)	-0.00617***	-0.0102***	0.0134***	-0.00234	0.0102***	-0.00236
	(0.00102)	(0.00115)	(0.00398)	(0.00458)	(0.00134)	(0.00398)
Issuer rating	0.00714***	-0.00255**	0.0252***	0.0145***	0.0306***	0.0176***
	(0.00113)	(0.00119)	(0.00382)	(0.00432)	(0.00117)	(0.00343)
Bond category FE	✓	✓	✓	✓	✓	✓
Fund x Year FE	✓	✓	✓	✓	✓	✓
Observations	1474793	1272599	114047	40908	1345830	138567
R-squared	-0.0211	-0.00411	-0.000120	-0.000953	0.00129	-0.00623

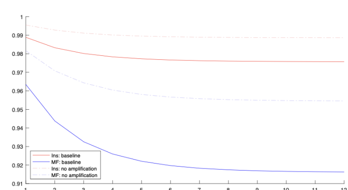
Note: This table shows the second stage estimates of demand on instrumented term-adjusted credit spreads. The instrument is constructed from equation (25). The outcome variable is the change in holdings relative to the index fund, normalized by the fund AUM: $\Delta\tilde{q}_{it}(n) = \Delta q_{it}(n) - \Delta q_{It}(n)$, where $\Delta q_{it}(n) = \ln\left(\frac{Q_{it}(n)}{A_{i,t}}\right) - \ln\left(\frac{Q_{it-1}(n)}{A_{i,t-1}}\right)$ and I is the largest index fund. Credit spreads are from the WRDS Bond Returns month-end transactions data, reported at the bond-quarter level. Controls include duration-matched US Treasury yield multiplied by the number of years remaining on the bond, the bid–ask spread as reported by WRDS, the number of years remaining, the initial amount issued (logged), and the issuer credit rating as reported in WRDS. The sample period is from 2007 to 2021. The first column reports the results for all life insurers, the second column reports results for property and casualty insurers, the third column reports results for ETFs, the fourth column reports results for index funds, the fifth column reports all active mutual funds, and the last column reports results for the residual investor. Index funds include only pure index funds whose objective is to match the investment performance of a specific securities market index. Includes bond type and fund–year fixed effects. Standard errors are clustered at the fund level.

5.3 Figure 1 and Figure 3: how the model works in equilibrium

Figure 1: Model dynamics: example



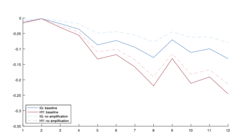
(a) Prices



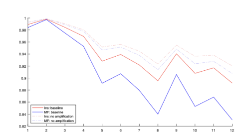
(b) AUM

Note: This graph shows simulated paths of AUM and asset prices given a series of smooth negative shocks. Parameters values are described in Section 3.2. "No amplification" refers to the case in which $\beta = 0$, meaning funds do not experience outflows in response to poor returns.

Figure 3: Model-implied dynamics



(a) Asset prices



(b) Fund AUM

Note: This graph shows the counterfactual AUM and asset prices with fundamental shocks to asset prices. There is no policy intervention in this simulation.

Read:

- Figure 1 is a stylized numerical example with two investor groups and two assets.
- Figure 3 applies the same logic to the March 2020 calibration and shows both asset prices and fund AUM under the baseline model and under a "no amplification" benchmark.
- In both figures, HY prices fall more than IG prices, and mutual fund AUM shrinks substantially more than insurer AUM.

Economic message. The dotted "no amplification" lines are crucial. They show directly how much of the decline comes from the endogenous feedback loop rather than from the original shock.

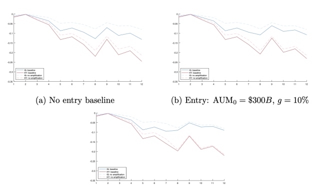
5.4 Figure 4: slow-moving arbitrage capital

Read:

- The paper lets an elastic arbitrageur enter after the crisis begins.
- With an initial AUM of \$300B and very high elasticity, the effect on prices is still modest.
- Prices only move close to the no-amplification benchmark when the entering arbitrageur is implausibly large, around \$2T in the illustrative simulation.

Economic message. Slow-moving arbitrage is not enough. The market is sufficiently dominated by large inelastic holders that even very elastic entrants need enormous balance sheets to offset amplification.

Figure 4: Slow-moving Arbitrage Capital: COVID crisis simulation



(a) No entry baseline

(b) Entry: $\text{AUM}_0 = \$300B$, $g = 10\%$

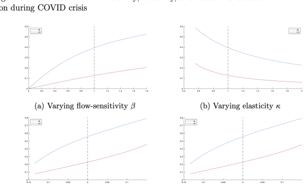
(c) Entry: $\text{AUM}_0 = \$2T$, $g = 10\%$

Table 4: Changing β and κ jointly

Scaling Factor for β	Scaling Factor for κ			
	0.8	1.0	1.2	1.4
0.6	34%	28%	24%	21%
0.8	41%	34%	29%	26%
1.0	46%	39%	34%	30%
1.2	51%	43%	38%	34%

Note: This table shows how flow-sensitivity and demand elasticity jointly affect amplification in our COVID crisis simulation. We conduct counterfactuals in which we scale the parameter vectors β and κ by different scaling factors. The scalar factor for β varies across rows. The scalar factor for κ varies across columns. Each cell displays a summary measure of amplification, defined as *Amplification* = (Price drop with amplification - Price drop without amplification)/ Price drop with amplification, where where price drops are computed over the last 12 days preceding the Fed announcement. The cell for which both scaling factors are 1 corresponds to the baseline estimate. We only show amplification for IG bonds.

Figure 5: The effects of flow-sensitivity, elasticity, and mutual fund sector size on amplification during COVID crisis



(a) Varying flow-sensitivity β

(b) Varying elasticity κ

(c) MF grows at expense of life insurers

(d) MF grows at expense of residual sector

Note: This figure shows how flow-sensitivity (Panel (a)), demand elasticity (Panel (b)), and the size of the mutual fund sector (Panels (c) and (d)) affect amplification in our COVID crisis simulation. We conduct counterfactuals in which we scale the parameter vector of interest (β or κ) by a scaling factor. The x-axis displays this scaling factor. The y-axis displays a summary measure of amplification, defined as *Amplification* = (Price drop with amplification - Price drop without amplification)/ Price drop with amplification, where where price drops are computed over the last 12 days preceding the Fed announcement. For reference, the dashed vertical line shows the baseline estimate (scaling factor of 1). We plot amplification separately for IG and HY bonds. In Panel (c), increase in MF bonds come from the life insurance sector, while in Panel (d) its comes from the residual sector.

5.5 Table 4 and Figure 5: what drives amplification

Read:

- In the baseline calibration, amplification for IG bonds is about 40%.
- Increasing flow sensitivity β raises amplification.
- Reducing demand elasticity κ raises amplification.
- Increasing the market share of mutual funds also raises amplification, regardless of whether their extra market share comes from insurers or the residual sector.

Economic message. Fragility is jointly determined by two forces:

- how fast investors redeem after bad performance,
- and how inelastic the remaining buyers are.

The larger the mutual fund sector becomes, the more dangerous the system becomes.

5.6 Table 5 and Figure 6: which funds hold which assets

Read:

- Funds with higher flow sensitivity hold smaller shares of long IG bonds and larger shares of short IG bonds.
- They also hold smaller shares of long HY bonds.
- Figure 6 shows a positive relationship between a fund's flow sensitivity and its demand elasticity among funds with downward-sloping demand.

Economic message. Flightier funds tilt toward assets that are easier to unload. This is intuitive from a portfolio-management perspective, but it also means some apparently liquid assets can become central nodes of systemic fragility because they are the first things funds sell.

Table 5: Portfolio shares regressed on flow-performance sensitivities

	(1) IG long share	(2) IG short share	(3) HY long share	(4) HY short share
beta, Winsorized fraction .05	-0.0302***	0.0413***	-0.0150***	0.00392***
	(0.00279)	(0.00211)	(0.00307)	(0.00100)
Retail share	-0.0254**	-0.0155*	0.0405***	0.000449
	(0.0122)	(0.00924)	(0.0134)	(0.00439)
Log AUM	0.00323	0.00294*	-0.00726***	0.00109
	(0.00221)	(0.00167)	(0.00243)	(0.000794)
Index MF indicator	0.295***	0.0645**	-0.302***	-0.0574***
	(0.0347)	(0.0263)	(0.0381)	(0.0125)
Constant	0.414***	0.126***	0.409***	0.0516***
	(0.0156)	(0.0118)	(0.0171)	(0.00562)
Year FE	✓	✓	✓	✓
Observations	6663	6663	6663	6663
R-squared	0.0322	0.0629	0.0241	0.0123

Note: This table shows results of regressing the market value of holdings for an asset class normalized by the total reported corporate bond holdings for each fund on fund characteristics. β is the fund-specific flow to performance sensitivity estimated using equation (23). Fund data is from CRSP Mutual Fund Holdings. Index fund indicator equals one for all funds whose objective is to match the investment performance of a specific securities market index. Includes year fixed effects. Fund-level beta estimates are winsorized at the 5% level.

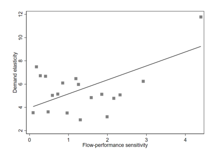


Figure 6: Funds with higher betas have higher elasticities

Note: This binned scatter plot compares the flow-to-performance sensitivity estimates to demand elasticities at the fund level. Includes funds with downward sloping estimated demand and flow performance sensitivities. Fund-level estimates include only those with at least 200 observations and are winsorized at 5%. The sign of the elasticity is flipped for this graph, so higher elasticities indicate more elastic funds.

Table 6: Systemic risk across assets

	IG		HY	
	Long	Short	Long	Short
Asset systemic measure 2019	1.47	1.60	1.68	1.50

Note: This table summarizes our measure of systemic importance of assets for year-end 2019. "IG" indicates bonds with credit rating of BBB- and above; "HY" indicates bonds with credit rating below BBB-. "Long" assets are those with five or more years remaining and "short" assets have fewer than 5 years remaining.

5.7 Table 6: systemic risk across assets

Read:

- IG long: 1.47
- IG short: 1.60
- HY long: 1.68
- HY short: 1.50

Economic message. Long HY bonds are the most systemic, but short IG bonds are more systemic than long IG bonds. This reverses the ranking one would get from a simple liquidity view. The paper's measure is about *who holds the asset in bad times*, not just about trading friction in isolation.

5.8 Figures 7–11: policy counterfactuals

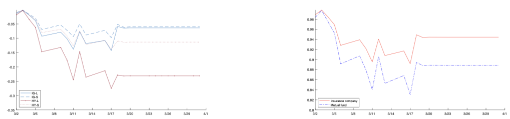
Figure 7: rate cut.

- A 50-basis-point policy-rate cut immediately raises bond prices and fund AUM.
- IG bonds rebound more than HY bonds because they have longer duration.

Figure 8: expected asset purchases.

- The central bank announces purchases of short-term IG bonds at $T = 14$ and purchases begin later.
- Short-term IG rebounds the most, but HY and long IG also recover because of positive contagion through fund balance sheets and mandates.

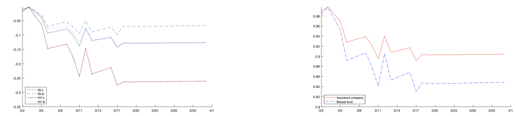
Figure 7: Counterfactual simulation: rate cut



(a) Prices with $T = 14$ (b) AUM with $T = 14$

Note: This graph shows the counterfactual AUM and asset prices following fundamental shocks to corporate bond prices. The central bank cut the policy rate by 50 basis points on day 14.

Figure 8: Counterfactual simulation: expected asset purchases



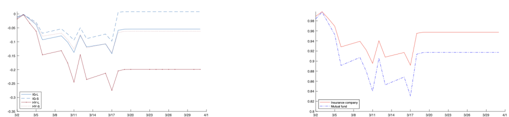
(a) Prices with $T = 14$ (b) AUM with $T = 14$

Note: This graph shows the counterfactual AUM and asset prices following fundamental shocks to corporate bond prices. The central bank plans asset purchases of 5% of short-term IG bonds at time $T = 20$, and announces the plan on day 14.

Figure 9: direct financing of mutual funds.

- The central bank transfers financing to mutual funds equal to 10% of the value of their IG holdings.
- This strongly supports both prices and fund AUM, and the gains spill over to insurers.

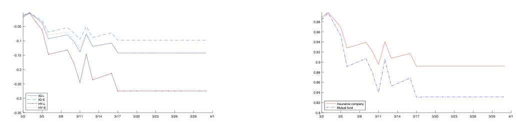
Figure 9: Counterfactual simulation: direct financing of mutual funds



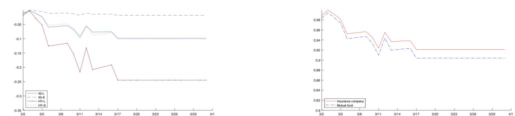
(a) Prices with $T = 14$ (b) AUM with $T = 14$

Note: This graph shows the counterfactual AUM and asset prices following fundamental shocks to corporate bond prices. The central bank transfers to all mutual funds 10% of the value of their IG holdings on day 14.

Figure 10: Counterfactual simulation: limits to redemption for mutual funds



(a) Prices with $T = 14$ (b) AUM with $T = 14$



(c) Prices with $T = 2$ (d) AUM with $T = 2$

Note: This graph shows the counterfactual AUM and asset prices following fundamental shocks to corporate bond prices. Mutual funds restrict suspend redemption on day 14 and day 2 for the upper and lower panels, respectively.

Figure 10: redemption restrictions.

- Freezing redemptions late is almost ineffective.
- Freezing them early is powerful because it prevents the flow spiral from getting started.

Figure 11: swing pricing.

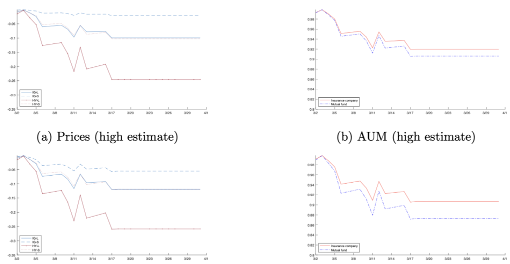
- Under the high estimate of swing pricing's effect, outflows are almost shut down and price declines are sharply reduced.
- Even under a more conservative calibration, the mitigation is still meaningful.

Economic message. The model helps separate policies that act through duration, policies that act through direct balance-sheet support, and policies that act by preventing the flow-performance loop itself.

5.9 Figure 12: which intervention has the highest bang for the buck?

Read:

Figure 11: Counterfactual simulation: swing pricing

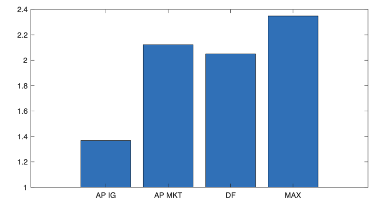


(a) Prices (high estimate) (b) AUM (high estimate)

(c) Prices (low estimate) (d) AUM (low estimate)

Note: This graph shows the counterfactual AUM and asset prices following fundamental shocks to corporate bond prices. Swing pricing is in place from $T = 0$. The top panel assumes a high effect of swing pricing, the bottom a lower effect.

Figure 12: Price impact multipliers of various interventions



Note: This graph shows the policy targeting multipliers of various interventions at $T = 14$ as described in Section 6. "AP IG" stands for expected asset purchases of 1% of ST IG bonds. "AP MKT" stands for expected asset purchases of 1% of all types of bonds. "DF" stands for direct financing of mutual funds. "MAX" stands for the maximum-price-impact benchmark.

- Expected purchases of short-term IG bonds have a multiplier around 1.4.
- Direct financing of mutual funds has a multiplier around 2.0.
- Broad market-wide asset purchases have a multiplier around 2.2.
- The theoretical maximum-price-impact benchmark is around 2.4.

Economic message. Direct financing is better targeted than narrow IG purchases because it directly supports flow-sensitive institutions. But broad asset purchases can be even closer to the optimum because they can target the sources of amplification across the whole market.

5.10 2015–2016 energy turmoil application

Read:

- The paper also applies the model to the oil-price-driven distress episode of 2015–2016.
- The stress is concentrated in energy and especially high-yield energy bonds.
- The model reproduces strong price declines and outflows in energy-focused funds with limited spillovers to non-energy sectors.

Economic message. This second application shows the framework is not specific to COVID. It can handle both system-wide crises and more localized sectoral shocks.

6 Short summary

- The paper builds a two-layer model in which households allocate across institutions and institutions allocate across assets.
- This structure makes it possible to quantify feedback between prices and flows, along with contagion across assets and institutions.
- Flow-to-performance estimates show that mutual fund outflows are very sensitive to bad returns.
- IV demand estimates show that corporate bond demand is downward sloping and that the residual sector is not nearly elastic enough to kill price pressure.

- In the March 2020 application, amplification is substantial, and slow-moving arbitrage capital is too small to undo it.
- Systemic fragility depends on who holds an asset, not just on the asset's own trading characteristics.
- Policy tools that directly weaken the flow spiral, such as direct financing or swing pricing, are especially powerful.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that nonbank fragility in corporate bond markets can be understood as the interaction of two forces: flow-sensitive household allocations into bond funds and inelastic asset demand by the institutions that hold the bonds. Once those two forces are jointly modeled, the system naturally generates amplification, cross-asset contagion, and cross-institution contagion. The model can be estimated from micro data, and it yields quantitatively meaningful policy counterfactuals.

7.2 Economic intuition

The key intuition is simple. Mutual funds promise daily liquidity to investors but hold relatively illiquid bonds. When prices fall, investors redeem. Funds then sell bonds to meet those redemptions, and because the set of buyers is not very elastic, prices fall again. What makes this especially powerful is that funds do not just sell the assets hit by the original shock; they also sell whatever is easiest or most convenient to sell. That is why even apparently safe or liquid assets can become systemic.

7.3 Takeaway

The message of the paper is not just that nonbanks matter. It is that the *combination* of flow-sensitive liabilities and inelastic asset demand creates a measurable network of fragility. Once that network is estimated, one can rank assets by systemic importance and compare policy tools by how well they target the amplification mechanism itself.